Static versus dynamic spring stiffness

EMAE 285 · Lab 5

Experiment overview

Tested three springs using static load–deflection and oscillation-based approaches. Examined how damping and natural frequency affect estimates of spring stiffness.

Variables and instrumentation

Load, displacement, period and damping assumptions

Analysis and presentation

Static and dynamic stiffness estimates; method comparison

Engineering relevance

Relevant to vibration isolation, machinery and component verification.

Scope and attribution

Course laboratory / group work unless noted. The resonance experiment is a proposed design and has no claimed measured outcomes.

Portfolio case-study brief based on coursework and user-provided descriptions. This is NOT a copy of the original laboratory report. Diagrams on the website are explanatory illustrations, not photographs or original experimental plots.

Original source (may require Google Drive permission):
https://docs.google.com/document/d/1Tp3VVJUmctt7nw34HLyBtWKbzEuUEX_6uWkXGyBkZec/edit